

code to it without the use of an external hardware programmer. It communicates using the STK500 protocol.

You can bypass the boot loader and program the microcontroller through in-circuit serial programming (ICSP) header, but using boot loader programming is quick and easy. Select the correct board from 'Tools->Board' menu in Arduino IDE 0022 and burn the program (sketch) through standard USB port in the computer.

We used Red Brick ESC module. If you use any other ESC module, you may have to modify delays in the code accordingly.

Programming an XBee

Several communication programs are available for programming an XBee, such as procontrol, xterm, pcu, X-CTU, even putty, but among these X-CTU is the easiest, Windows-based graphical program, though it takes longer time. But once you master X-CTU, you can quickly connect XBee to any other communication program and do the reprogramming of your XBee through short AT commands.

Download X-CTU from the Internet and install it on a Windows PC. Position the XBee on the Explorer board and connect it to the USB port of your PC. Fire up X-CTU and see whether it detects the device or not.

The two XBees can be made to communicate with each other freely and precisely through X-CTU software. The X-CTU has four tabbed windows: PC settings, Range test, Modem and Configuration. In PC settings window there are three sub-windows located at the bottom: Host setup, User COM ports and Network Interface. In case you have not found your right COM ports, you can click Network Interface and then Digi Device Discovery to find your device.

Press Test/Query button on right side of the same PC Settings window and see what XBee it discovers. Check Enable API box because we will use this feature in Coordinator XBee (XB1). Now come to Configuration window where most of the settings need to be

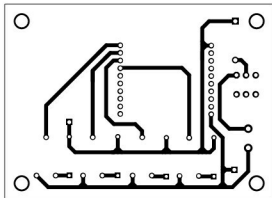


Fig. 7: An actual-size PCB pattern for XBee-controlled aircraft (transmitter side)

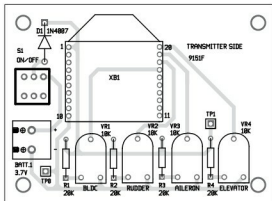


Fig. 8: Component layout for the PCB (transmitter side)

made.

In Configuration window, click on Read button and your screen will start showing lots of data. Do not get alarmed; you need to configure only a few parameters.

Click on Always Update Firmware, as this will keep the XBee updated with the latest software. In Function Set pull-down menu select XBee Coordinator API. In Version pull-down select the highest number. In Networking parameters window find the personal area network (PAN) parameter, and give it a 4-digit number, say, 7821. Do not give 1234 because most people use this number. In case another person is experimenting with XBee in your area, you both will collide in the air space for bandwidth.

For communication between two XBees, one has to be made Coordinator and the other a Router. Here we are preparing the Coordinator XBee (XB1).

Coming to the Addressing param-

eter at DH and DL, these are called the High and Low addresses. DH is a 32-bit address written on the XBee itself. Enter this parameter here. In most cases, it is the same number for all generic XBees: 0013A200 (find out yours to be sure). Just below this number you will find another 32-bit number inscribed on the XBee. This is the DL number.

The DL parameter is to be switched between the Coordinator and the Router; the Router number to be entered into the Coordinator and vice versa. After doing that check Write and the setup will be written in your Coordinator XBee. See the message below. You may need your Internet connection on now. When it is over, 'finished' message will appear at the bottom.

Pat yours back; you have set up the XBee Coordinator successfully! Now remove the Coordinator and insert the other XBee (XB2) and open Configuration window. After checking Read, and getting the parameters shown on the middle window, go to the Function Set pull-down menu and select XBee Router API. In Version pull-down select the highest number. Find the PAN in the Networking parameter and enter 7821 (same as the Coordinator XBee). Set the JV parameter as 1, as it will set the Router to rejoin the coordinator on startup.

In Addressing parameter the DH shall be as written on the XBee and the DL shall be the DL number written on the Coordinator XBee. So far so good! Now come down further to IO Settings to set up analogue and digital data entry part of the router XBee.

In the IO Settings parameter, D0, D1, D2, D3 are the four analogue input pins through which we will send our 4-channel analogue data. By default these pins are disabled. To enable them, go to IO Settings and make D0,